Exercises: Probability — Basics – Answer Key

Mathematical Foundations for Machine Learning

Discrete Probability and Distributions

Exercise 1.1 [Easy]

A fair six-sided die is rolled once. Let X be the random variable defined as

$$X = \begin{cases} 1 & \text{if the outcome is even,} \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find the probability mass function P_X.

(b) Compute $\mathbb{E}[X]$ and $\mathrm{Var}[X]$.

Solution

(a) The even outcomes are $\{2, 4, 6\}$ and the odd outcomes are $\{1, 3, 5\}$. Each has probability $3/6 = 1/2$, so

$$P_X(X = 1) = \frac{1}{2}, \qquad P_X(X = 0) = \frac{1}{2}.$$

This is a Bernoulli distribution with parameter $p = 1/2$.

(b)

$$\mathbb{E}[X] = 0 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2} = \frac{1}{2}.$$

Since $X^2 = X$ (as $X \in \{0, 1\}$), we have $\mathbb{E}[X^2] = 1/2$, so

$$\mathrm{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}.$$

Exercise 1.2 [Medium]

A biased coin has probability p of showing heads. It is tossed n times independently. Let X denote the number of heads observed.

(a) Write down the probability mass function of X and verify that the probabilities sum to 1.

(b) Using the binomial distribution, show that $\mathbb{E}[X] = np$.

(c) A machine learning classifier makes independent predictions on 20 test samples, each correct with probability 0.85. What is the probability that exactly 17 predictions are correct?

Solution

(a) $X \sim \mathrm{Bin}(n, p)$ with

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, \ldots, n.$$

Summing over all k:

$$\sum_{k=0}^{n} \binom{n}{k} p^k (1-p)^{n-k} = (p + (1-p))^n = 1^n = 1,$$

by the binomial theorem.

(b)

$$\mathbb{E}[X] = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k}.$$

The $k = 0$ term vanishes. For $k \geq 1$, use $k\binom{n}{k} = n\binom{n-1}{k-1}$:

$$\mathbb{E}[X] = np \sum_{k=1}^{n} \binom{n-1}{k-1} p^{k-1}(1-p)^{n-k} = np \sum_{j=0}^{n-1} \binom{n-1}{j} p^j (1-p)^{n-1-j} = np.$$

(c) Here $n = 20$, $p = 0.85$, $k = 17$:

$$P(X = 17) = \binom{20}{17} (0.85)^{17} (0.15)^3 = 1140 \cdot (0.85)^{17} \cdot (0.15)^3 \approx 0.243.$$

Exercise 1.3 [Hard]

Consider a discrete probability space $(\Omega, 2^\Omega, P)$ with $\Omega = \{\omega_1, \omega_2, \omega_3\}$ and $P(\omega_i) = p_i > 0$ for $i = 1, 2, 3$.

(a) How many events does 2^Ω contain? Verify that the additivity rule $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ holds for $A = \{\omega_1, \omega_2\}$ and $B = \{\omega_2, \omega_3\}$.

(b) Define random variables $X, Y : \Omega \to \mathbb{R}$ by $X(\omega_i) = i$ and $Y(\omega_i) = i^2$. Compute $\mathbb{E}[XY]$ in terms of p_1, p_2, p_3.

(c) When moving from a finite Ω to an uncountable one like $\mathbb{R}$, the powerset 2^Ω must be replaced by a σ-algebra. Give the three defining properties of a σ-algebra and briefly explain why this restriction is necessary.

Solution

(a) $|2^\Omega| = 2^3 = 8$. We have

$$\begin{aligned} P(A) &= p_1 + p_2, \quad P(B) = p_2 + p_3, \\ A \cap B &= \{\omega_2\}, \quad P(A \cap B) = p_2, \\ A \cup B &= \{\omega_1, \omega_2, \omega_3\} = \Omega, \quad P(A \cup B) = 1. \end{aligned}$$

Check: $P(A) + P(B) - P(A \cap B) = (p_1 + p_2) + (p_2 + p_3) - p_2 = p_1 + p_2 + p_3 = 1$. ✓

(b)

$$\mathbb{E}[XY] = \sum_{i=1}^{3} X(\omega_i) Y(\omega_i)\, p_i = 1 \cdot 1 \cdot p_1 + 2 \cdot 4 \cdot p_2 + 3 \cdot 9 \cdot p_3 = p_1 + 8p_2 + 27p_3.$$

(c) A σ-algebra $\mathcal{A} \subseteq 2^\Omega$ satisfies:

1. $\emptyset \in \mathcal{A}$ and $\Omega \in \mathcal{A}$,
2. If $A \in \mathcal{A}$, then $A^c \in \mathcal{A}$ (closed under complements),
3. If $A_1, A_2, \ldots \in \mathcal{A}$, then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{A}$ and $\bigcap_{i=1}^{\infty} A_i \in \mathcal{A}$ (closed under countable unions and intersections).

This restriction is necessary because assigning probabilities consistently to *all* subsets of an uncountable set leads to paradoxes (e.g., the Banach–Tarski paradox shows one can decompose a ball into pieces and reassemble them into two balls of the same volume). By restricting to a σ-algebra, we only assign probabilities to "well-behaved" (measurable) sets.

Continuous Distributions and Density Functions

Exercise 2.1 [Easy]

Consider the function

$$p(x) = \begin{cases} c\,(1-x^2) & \text{if } x \in [-1,1], \\ 0 & \text{otherwise.} \end{cases}$$

(a) Find the constant $c > 0$ such that p is a valid probability density.

(b) Compute the cumulative distribution function $F(x) = P((-\infty, x])$ for $x \in [-1,1]$.

Solution

(a) We need $\int_{-1}^{1} c(1-x^2)\,dx = 1$. Compute:

$$\int_{-1}^{1} (1-x^2)\,dx = \left[x - \frac{x^3}{3}\right]_{-1}^{1} = \left(1 - \frac{1}{3}\right) - \left(-1 + \frac{1}{3}\right) = \frac{4}{3}.$$

Thus $c = 3/4$.

(b) For $x \in [-1,1]$:

$$F(x) = \int_{-1}^{x} \frac{3}{4}(1-t^2)\,dt = \frac{3}{4}\left[t - \frac{t^3}{3}\right]_{-1}^{x} = \frac{3}{4}\left(x - \frac{x^3}{3} + 1 - \frac{1}{3}\right) = \frac{3}{4}\left(x - \frac{x^3}{3} + \frac{2}{3}\right).$$

For $x < -1$: $F(x) = 0$. For $x > 1$: $F(x) = 1$.

Exercise 2.2 [Medium]

Let $X \sim \mathcal{N}(\mu, \sigma^2)$ be a univariate Gaussian random variable with density

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

(a) Show that if p is symmetric around μ, then the mean and the median of X coincide.

(b) The α-quantile q_α satisfies $F(q_\alpha) = \alpha$. Using symmetry of p around μ, show that $q_\alpha = 2\mu - q_{1-\alpha}$.

(c) In a machine learning context, a model's prediction error follows $\mathcal{N}(0,4)$. Find an expression (in terms of q_α) for the symmetric interval $[-a, a]$ that contains 95% of the errors.

Solution

(a) A density symmetric around μ satisfies $p(\mu + t) = p(\mu - t)$ for all t. The median m satisfies $P(X \le m) = 1/2$. By symmetry,

$$P(X \le \mu) = \int_{-\infty}^{\mu} p(x)\,dx = \int_{0}^{\infty} p(\mu - t)\,dt = \int_{0}^{\infty} p(\mu + t)\,dt = P(X \ge \mu).$$

Since $P(X \le \mu) + P(X \ge \mu) = 1$, we get $P(X \le \mu) = 1/2$, so $m = \mu = \mathbb{E}[X]$.

(b) By symmetry, $F(\mu + t) = 1 - F(\mu - t)$ for all t. Let $q_\alpha = \mu + t_\alpha$, so $F(\mu + t_\alpha) = \alpha$. Then

$$F(\mu - t_\alpha) = 1 - \alpha,$$

which means $q_{1-\alpha} = \mu - t_\alpha$. Adding: $q_\alpha + q_{1-\alpha} = 2\mu$, hence $q_\alpha = 2\mu - q_{1-\alpha}$.

(c) We need $P(-a \le X \le a) = 0.95$. By symmetry around $\mu = 0$:

$$P(X \le a) - P(X \le -a) = 0.95.$$

Since $P(X \leq -a) = 1 - P(X \leq a)$ by symmetry, this gives $2P(X \leq a) - 1 = 0.95$, so $P(X \leq a) = 0.975$. Thus $a = q_{0.975}$. For $\mathcal{N}(0, 4)$ with $\sigma = 2$, we have $a = 2\, q^{\text{std}}_{0.975}$ where $q^{\text{std}}_{0.975} \approx 1.96$ is the quantile of the standard normal. So $a \approx 3.92$.

Exercise 2.3 [Hard]

Let $X \sim \mathcal{N}(\mu, C)$ be a $\mathbb{R}^d$-valued random variable with density

$$p(x) = \frac{1}{\sqrt{(2\pi)^d \det C}} \exp\left(-\frac{1}{2}(x-\mu)^T C^{-1}(x-\mu)\right),$$

where $C \in \mathbb{R}^{d\times d}$ is symmetric and positive definite.

(a) Show that the density $p(x)$ is constant on the ellipsoid $\{x \in \mathbb{R}^d : (x-\mu)^T C^{-1}(x-\mu) = r^2\}$ for any fixed $r > 0$.

(b) Let $C = \sigma^2 I$ (isotropic case). Show that the density depends on x only through $\|x-\mu\|_2$, and hence the level sets are spheres.

(c) Let $v \in \mathbb{R}^d$ be a fixed unit vector. Define the scalar random variable $Z = v^T X$. Using the properties of expectation and the covariance matrix, show that $\mathbb{E}[Z] = v^T\mu$ and $\mathrm{Var}(Z) = v^T C v$.

Solution

(a) For any x on the ellipsoid, $(x-\mu)^T C^{-1}(x-\mu) = r^2$. Thus

$$p(x) = \frac{1}{\sqrt{(2\pi)^d \det C}}\, e^{-r^2/2},$$

which is the same constant for all points on the ellipsoid.

(b) When $C = \sigma^2 I$, we have $C^{-1} = \frac{1}{\sigma^2} I$ and $\det C = \sigma^{2d}$. The exponent becomes

$$(x-\mu)^T \frac{1}{\sigma^2} I\, (x-\mu) = \frac{\|x-\mu\|_2^2}{\sigma^2}.$$

Thus $p(x) = \frac{1}{(2\pi\sigma^2)^{d/2}} \exp\left(-\frac{\|x-\mu\|_2^2}{2\sigma^2}\right)$, which depends on x only through $\|x-\mu\|_2$. The level sets $\{x : \|x-\mu\|_2 = r\}$ are spheres centered at μ.

(c) By linearity of expectation:

$$\mathbb{E}[Z] = \mathbb{E}[v^T X] = v^T \mathbb{E}[X] = v^T\mu.$$

For the variance, using $\mathrm{Var}(Z) = \mathbb{E}[(Z - \mathbb{E}[Z])^2]$:

$$Z - \mathbb{E}[Z] = v^T(X-\mu),$$

so

$$\mathrm{Var}(Z) = \mathbb{E}[(v^T(X-\mu))^2] = \mathbb{E}[v^T(X-\mu)(X-\mu)^T v] = v^T\mathbb{E}[(X-\mu)(X-\mu)^T]v = v^T C v.$$

Joint Distributions, Marginals, and Transformations

Exercise 3.1 [Easy]

Let (X_1, X_2) be a random variable in $\mathbb{R}^2$ with joint density

$$p(x_1, x_2) = \begin{cases} 6\,x_1\,x_2^2 & \text{if } 0 < x_1 < 1,\ 0 < x_2 < 1, \\ 0 & \text{otherwise.} \end{cases}$$

(a) Verify that p is a valid joint density.

(b) Compute the marginal densities $p_{X_1}(x_1)$ and $p_{X_2}(x_2)$.

Solution

(a)

$$\int_0^1 \int_0^1 6\,x_1\,x_2^2\,dx_2\,dx_1 = 6 \int_0^1 x_1\,dx_1 \int_0^1 x_2^2\,dx_2 = 6 \cdot \frac{1}{2} \cdot \frac{1}{3} = 1. \quad \checkmark$$

(b) Marginal of X_1:

$$p_{X_1}(x_1) = \int_0^1 6\,x_1\,x_2^2\,dx_2 = 6\,x_1 \cdot \frac{1}{3} = 2\,x_1, \quad x_1 \in (0,1).$$

Marginal of X_2:

$$p_{X_2}(x_2) = \int_0^1 6\,x_1\,x_2^2\,dx_1 = 6\,x_2^2 \cdot \frac{1}{2} = 3\,x_2^2, \quad x_2 \in (0,1).$$

Exercise 3.2 [Medium]

Suppose U is uniformly distributed on $[0,1]$, i.e. $p_U(u) = 1$ for $u \in [0,1]$.

(a) Using the transformation law, find the density of $Y = -\frac{1}{\lambda}\log(U)$ for $\lambda > 0$. What well-known distribution does Y follow?

(b) Now let $Z = U^2$. Using the transformation law, find the density $p_Z(z)$.

(c) In the context of sampling: suppose you only have access to a uniform random number generator. Explain how you would use part (a) to generate samples from an exponential distribution with rate $\lambda = 2$.

Solution

(a) The transformation is $g(u) = -\frac{1}{\lambda}\log(u)$, which maps $(0,1] \to [0,\infty)$. The inverse is $g^{-1}(y) = e^{-\lambda y}$. Its derivative is

$$\frac{dg^{-1}}{dy} = -\lambda\,e^{-\lambda y}.$$

By the transformation law:

$$p_Y(y) = p_U(g^{-1}(y)) \left| \frac{dg^{-1}}{dy} \right| = 1 \cdot \lambda\,e^{-\lambda y} = \lambda\,e^{-\lambda y}, \quad y \geq 0.$$

This is the exponential distribution with rate λ.

(b) Let $g(u) = u^2$, mapping $(0,1) \to (0,1)$. The inverse is $g^{-1}(z) = \sqrt{z}$ with derivative $\frac{1}{2\sqrt{z}}$. By the transformation law:

$$p_Z(z) = p_U(\sqrt{z}) \left| \frac{1}{2\sqrt{z}} \right| = \frac{1}{2\sqrt{z}}, \quad z \in (0,1).$$

(c) Draw a sample u from Uniform$(0,1)$. Compute $y = -\frac{1}{2}\log(u)$. By part (a), y is a sample from the exponential distribution with rate $\lambda = 2$, i.e. $p_Y(y) = 2e^{-2y}$ for $y \geq 0$.

Exercise 3.3 [Hard]

Let X be a $\mathbb{R}^d$-valued random variable with density p_X, and let $g : \mathbb{R}^d \to \mathbb{R}^d$ be a continuously differentiable bijection with non-vanishing Jacobian determinant.

(a) State the transformation law: write the density p_Y of $Y = g(X)$ in terms of p_X, g^{-1}, and the Jacobian of g^{-1}.

(b) Let $d = 2$, $X \sim \mathcal{N}(0, I_2)$, and define $g(x_1, x_2) = (2x_1 + x_2,\ x_2)$. Compute the Jacobian matrix $Dg(x)$, its determinant, and find the density $p_Y(y)$.

(c) In the setting of **(b)**, is Y still Gaussian? If so, find its mean and covariance matrix C_Y directly from the linear transformation $Y = AX$ and verify it matches the density you found.

Solution

(a) The transformation law states:

$$p_Y(y) = p_X\big(g^{-1}(y)\big)\,\big|\det Dg^{-1}(y)\big|,$$

where $Dg^{-1}(y)$ is the Jacobian matrix of g^{-1} at y, with entries $(Dg^{-1})_{ij} = \frac{\partial (g^{-1})_i}{\partial y_j}$.

(b) The Jacobian of g is

$$Dg = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}, \qquad \det Dg = 2.$$

The inverse transformation is $g^{-1}(y_1, y_2) = \big(\frac{y_1 - y_2}{2},\ y_2\big)$ with

$$Dg^{-1} = \begin{pmatrix} 1/2 & -1/2 \\ 0 & 1 \end{pmatrix}, \qquad |\det Dg^{-1}| = \frac{1}{2}.$$

Since $p_X(x) = \frac{1}{2\pi} e^{-\|x\|^2/2}$:

$$p_Y(y) = \frac{1}{2\pi} \exp\left(-\frac{1}{2}\left(\frac{(y_1 - y_2)^2}{4} + y_2^2\right)\right) \cdot \frac{1}{2} = \frac{1}{4\pi} \exp\left(-\frac{(y_1 - y_2)^2 + 4y_2^2}{8}\right).$$

(c) Yes, a linear transformation of a Gaussian is Gaussian. Here $Y = AX$ with $A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$. Since $\mathbb{E}[X] = 0$:

$$\mathbb{E}[Y] = A\,\mathbb{E}[X] = 0, \qquad C_Y = AI_2A^T = AA^T = \begin{pmatrix} 5 & 1 \\ 1 & 1 \end{pmatrix}.$$

The Gaussian density with this covariance is $p_Y(y) = \frac{1}{2\pi\sqrt{\det C_Y}} \exp\big(-\frac{1}{2} y^T C_Y^{-1} y\big)$. We have $\det C_Y = 4$ and $C_Y^{-1} = \frac{1}{4}\begin{pmatrix} 1 & -1 \\ -1 & 5 \end{pmatrix}$. Expanding $y^T C_Y^{-1} y = \frac{1}{4}(y_1^2 - 2y_1y_2 + 5y_2^2) = \frac{(y_1 - y_2)^2 + 4y_2^2}{4}$. So

$$p_Y(y) = \frac{1}{4\pi} \exp\left(-\frac{(y_1 - y_2)^2 + 4y_2^2}{8}\right),$$

which matches part (b). ✓

Expectation, Variance, Covariance, and Correlation

Exercise 4.1 [Easy]

Let X and Y be two real-valued random variables with $\mathbb{E}[X] = 2$, $\mathbb{E}[Y] = -1$, $\mathrm{Var}(X) = 3$, $\mathrm{Var}(Y) = 5$, and $\mathrm{Cov}(X, Y) = -2$.

(a) Compute $\mathrm{Var}(X + Y)$ and $\mathrm{Var}(2X - 3Y + 7)$.

(b) Compute $\mathrm{Corr}(X, Y)$ and determine whether X and Y are positively or negatively correlated.

Solution

(a) Using $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\mathrm{Cov}(X, Y)$:

$$\mathrm{Var}(X + Y) = 3 + 5 + 2(-2) = 4.$$

For $\mathrm{Var}(2X - 3Y + 7)$: constants do not affect variance, so

$$\mathrm{Var}(2X - 3Y) = 4\,\mathrm{Var}(X) + 9\,\mathrm{Var}(Y) + 2(2)(-3)\mathrm{Cov}(X, Y) = 12 + 45 + 24 = 81.$$

(Using $\mathrm{Var}(aX + bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X, Y)$ with $a = 2, b = -3$.)
So $\mathrm{Var}(2X - 3Y + 7) = 81$.

(b)

$$\mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sigma(X)\sigma(Y)} = \frac{-2}{\sqrt{3} \cdot \sqrt{5}} = \frac{-2}{\sqrt{15}} \approx -0.516.$$

Since the correlation is negative, X and Y are negatively correlated.

Exercise 4.2 [Medium]

A coin has probability p for heads. It is tossed twice independently. Let X denote the result of the first toss and Y the result of the second toss (1 for heads, 0 otherwise).

(a) Compute $\mathbb{E}[XY]$. *(cf. Exam 2021, Ex. 4b)*

(b) Compute $\mathrm{Cov}(X, Y)$ and explain the result using independence.

(c) Now define $Z = X + Y$ (total number of heads). Compute $\mathrm{Cov}(X, Z)$ and $\mathrm{Corr}(X, Z)$.

Solution

(a) Since X and Y are independent:

$$\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y] = p \cdot p = p^2.$$

(b)

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = p^2 - p \cdot p = 0.$$

This is expected: independent random variables always have zero covariance. (Note: the converse is not true in general.)

(c) Since $Z = X + Y$:

$$\mathrm{Cov}(X, Z) = \mathrm{Cov}(X, X + Y) = \mathrm{Cov}(X, X) + \mathrm{Cov}(X, Y) = \mathrm{Var}(X) + 0 = p(1 - p).$$

For the correlation: $\mathrm{Var}(Z) = \mathrm{Var}(X) + \mathrm{Var}(Y) = 2p(1 - p)$, so

$$\mathrm{Corr}(X, Z) = \frac{p(1 - p)}{\sqrt{p(1 - p)}\,\sqrt{2p(1 - p)}} = \frac{1}{\sqrt{2}}.$$

Notably, this does not depend on p (as long as $p \in (0, 1)$).

Exercise 4.3 [Hard]

Let X be a real-valued random variable with a symmetric density, i.e. $p(x) = p(-x)$ for all x, and let $Y = X^2$.

(a) Show that $\mathrm{Cov}(X, Y) = 0$.

(b) Despite zero covariance, X and Y are clearly dependent (since Y is a function of X). Does $\mathrm{Corr}(X, Y) = 0$ imply independence? Justify your answer.

(c) Now let $X \sim \mathcal{N}(0, \sigma^2 I_d)$ be a d-dimensional Gaussian and $A \in \mathbb{R}^{d\times d}$ symmetric. Show that

$$\mathbb{E}[\langle X, AX\rangle] = \sigma^2 \operatorname{tr}(A).$$

Solution

(a) By symmetry of p, $\mathbb{E}[X] = 0$ and $\mathbb{E}[X^3] = \int x^3 p(x)\,dx = 0$ (since $x^3 p(x)$ is an odd function integrated over a symmetric domain). Thus

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \mathbb{E}[X^3] - 0 \cdot \mathbb{E}[X^2] = 0.$$

(b) No. Correlation measures *linear* dependence only. X and $Y = X^2$ are a classic counterexample: Y is completely determined by X (hence maximally dependent), yet $\mathrm{Corr}(X, Y) = 0$. Zero correlation does **not** imply independence in general. (It does if (X, Y) is jointly Gaussian, but that is a special case.)

(c) Expand the quadratic form:

$$\langle X, AX\rangle = \sum_{i=1}^{d}\sum_{j=1}^{d} a_{ij} X_i X_j.$$

Taking expectation and using $\mathbb{E}[X_i X_j] = \sigma^2 \delta_{ij}$ (since $X \sim \mathcal{N}(0, \sigma^2 I)$):

$$\mathbb{E}[\langle X, AX\rangle] = \sum_{i,j} a_{ij}\,\mathbb{E}[X_i X_j] = \sum_{i,j} a_{ij}\,\sigma^2 \delta_{ij} = \sigma^2 \sum_{i=1}^{d} a_{ii} = \sigma^2 \operatorname{tr}(A).$$

Independence, Conditional Probability, and Bayes' Theorem

Exercise 5.1 [Easy]

Let (X_1, X_2) have joint density $p(x_1, x_2) = 6\,x_1\,x_2^2$ on $[0, 1]^2$ (as in Exercise 3.1).

(a) Using the marginal densities $p_{X_1}(x_1) = 2x_1$ and $p_{X_2}(x_2) = 3x_2^2$, determine whether X_1 and X_2 are independent.

(b) Compute $\mathbb{E}[X_1 X_2]$ and verify that it equals $\mathbb{E}[X_1]\,\mathbb{E}[X_2]$.

Solution

(a) Check: $p_{X_1}(x_1)\,p_{X_2}(x_2) = 2x_1 \cdot 3x_2^2 = 6x_1 x_2^2 = p(x_1, x_2)$ for all $(x_1, x_2) \in [0, 1]^2$. Since the joint density factors as the product of the marginals, X_1 and X_2 are **independent**.

(b)

$$\mathbb{E}[X_1 X_2] = \int_0^1 \int_0^1 x_1 x_2 \cdot 6x_1 x_2^2\,dx_2\,dx_1 = 6 \int_0^1 x_1^2\,dx_1 \int_0^1 x_2^3\,dx_2 = 6 \cdot \frac{1}{3} \cdot \frac{1}{4} = \frac{1}{2}.$$

Separately: $\mathbb{E}[X_1] = \int_0^1 2x_1^2\,dx_1 = \frac{2}{3}$ and $\mathbb{E}[X_2] = \int_0^1 3x_2^3\,dx_2 = \frac{3}{4}$.

Check: $\mathbb{E}[X_1]\,\mathbb{E}[X_2] = \frac{2}{3} \cdot \frac{3}{4} = \frac{1}{2} = \mathbb{E}[X_1 X_2]$. ✓
This is consistent with independence.

Exercise 5.2 [Medium]

A medical test for a disease has the following properties: if a patient has the disease, the test is positive with probability 0.95 (sensitivity). If the patient is healthy, the test is positive with probability 0.03 (false positive rate). The prevalence of the disease in the population is 0.01.

(a) Using Bayes' theorem, compute the probability that a patient actually has the disease given a positive test result.

(b) The result of part (a) is often surprisingly low. Using the law of total probability, explain intuitively why this happens by computing $P(\text{positive test})$.

(c) At what prevalence level π would a positive test result give a 50% probability of having the disease? *(cf. Exam 2021, Ex. 4c)*

Solution

(a) Let D = has disease, T^+ = test positive. We are given $P(T^+|D) = 0.95$, $P(T^+|D^c) = 0.03$, $P(D) = 0.01$. By Bayes' theorem:

$$P(D|T^+) = \frac{P(T^+|D)\,P(D)}{P(T^+)} = \frac{0.95 \times 0.01}{P(T^+)}.$$

(b) By the law of total probability (using the partition $\{D, D^c\}$):

$$P(T^+) = P(T^+|D)P(D) + P(T^+|D^c)P(D^c) = 0.95 \cdot 0.01 + 0.03 \cdot 0.99 = 0.0095 + 0.0297 = 0.0392.$$

Thus $P(D|T^+) = \frac{0.0095}{0.0392} \approx 0.242$, i.e. only about 24%.
The reason: the disease is rare (1% prevalence), so the large healthy population generates many false positives ($0.03 \times 0.99 = 0.0297$), which outnumber the true positives ($0.95 \times 0.01 = 0.0095$).

(c) We want $P(D|T^+) = 0.5$. Setting up:

$$0.5 = \frac{0.95\,\pi}{0.95\,\pi + 0.03(1-\pi)}.$$

Solving: $0.95\pi + 0.03(1-\pi) = 2 \times 0.95\pi$, so $0.03 - 0.03\pi = 0.95\pi$, giving $0.03 = 0.98\pi$, hence $\pi = \frac{0.03}{0.98} \approx 0.0306$, i.e. about 3.06% prevalence.

Exercise 5.3 [Hard]

Let X and Y be real-valued random variables with joint density $p(x, y)$.

(a) Write down the conditional density $p(x|y)$ and verify that $\int_{\mathbb{R}} p(x|y)\,dx = 1$ for any y with $p(y) > 0$.

(b) Prove the tower property of conditional expectation: $\mathbb{E}[\mathbb{E}[X|Y]] = \mathbb{E}[X]$.

(c) In a mixture of discrete and continuous spaces: let $Y \in \{1, \ldots, K\}$ be a class label and $X \in \mathbb{R}^d$ a feature vector with joint distribution $p(x, k)$. Show that the posterior $P(Y = k|x) = \frac{p(x|Y=k)\,P(Y=k)}{\sum_{l=1}^{K} p(x|Y=l)\,P(Y=l)}$ defines a valid probability measure over $\{1, \ldots, K\}$ for each x.

Solution

(a) The conditional density is defined as

$$p(x|y) = \frac{p(x,y)}{p(y)}, \quad \text{where } p(y) = \int_{\mathbb{R}} p(x,y)\,dx > 0.$$

Verification:

$$\int_{\mathbb{R}} p(x|y)\,dx = \int_{\mathbb{R}} \frac{p(x,y)}{p(y)}\,dx = \frac{1}{p(y)} \int_{\mathbb{R}} p(x,y)\,dx = \frac{p(y)}{p(y)} = 1.$$

(b) $\mathbb{E}[X|Y]$ is a function of Y, say $h(Y) = \mathbb{E}[X|Y=y]$. Then:

$$\begin{aligned}
\mathbb{E}[\mathbb{E}[X|Y]] &= \int_{\mathbb{R}} h(y)\,p(y)\,dy = \int_{\mathbb{R}} \left(\int_{\mathbb{R}} x\,p(x|y)\,dx \right) p(y)\,dy \\
&= \int_{\mathbb{R}} \int_{\mathbb{R}} x\,\frac{p(x,y)}{p(y)}\,p(y)\,dx\,dy = \int_{\mathbb{R}} \int_{\mathbb{R}} x\,p(x,y)\,dx\,dy \\
&= \int_{\mathbb{R}} x\,p(x)\,dx = \mathbb{E}[X].
\end{aligned}$$

(c) We need to check: (i) $P(Y=k|x) \geq 0$, and (ii) $\sum_{k=1}^{K} P(Y=k|x) = 1$.
(i) Since $p(x|Y=k) \geq 0$ and $P(Y=k) \geq 0$, the numerator is non-negative, and the denominator is positive (assuming $p(x) = \sum_l p(x|Y=l)P(Y=l) > 0$).
(ii) Summing over k:

$$\sum_{k=1}^{K} P(Y=k|x) = \sum_{k=1}^{K} \frac{p(x|Y=k)P(Y=k)}{\sum_{l=1}^{K} p(x|Y=l)P(Y=l)} = \frac{\sum_{k=1}^{K} p(x|Y=k)P(Y=k)}{\sum_{l=1}^{K} p(x|Y=l)P(Y=l)} = 1.$$

Thus $P(Y=\cdot\,|x)$ defines a valid probability measure over $\{1,\ldots,K\}$.

Mixed Challenging Exercises

Exercise 6.1 [Hard]

Consider N data points $\{(x_n, y_n)\}_{n=1}^{N}$ with $x_n \in \mathbb{R}^D$ and $y_n \in \mathbb{R}$, generated by the model $y = x^T\theta + \varepsilon$ where $\varepsilon \sim \mathcal{N}(0,\sigma^2)$ i.i.d.

(a) Write the conditional density $p(y_n|x_n;\theta)$ for a single observation.

(b) Using the i.i.d. assumption, write the joint conditional density $P(Y|X;\theta)$ for all N observations (where $X \in \mathbb{R}^{N\times D}$ is the design matrix and $Y \in \mathbb{R}^N$ the target vector).

(c) Show that the maximum likelihood estimate of θ is $\hat{\theta} = (X^TX)^{-1}X^TY$.

(d) Show that maximizing the log-likelihood is equivalent to minimizing $\|Y - X\theta\|_2^2$. Explain the connection: under what assumption on the noise does least-squares regression correspond to maximum likelihood?

(cf. Exam 2021, Ex. 4d)

Solution

(a) Since $y_n = x_n^T\theta + \varepsilon_n$ with $\varepsilon_n \sim \mathcal{N}(0,\sigma^2)$, we have $y_n|x_n \sim \mathcal{N}(x_n^T\theta, \sigma^2)$:

$$p(y_n|x_n;\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_n - x_n^T\theta)^2}{2\sigma^2} \right).$$

(b) By independence:

$$P(Y|X;\theta) = \prod_{n=1}^{N} p(y_n|x_n;\theta) = \prod_{n=1}^{N} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y_n - x_n^T\theta)^2}{2\sigma^2}\right).$$

(c) The log-likelihood is:

$$\ell(\theta) = \log P(Y|X;\theta) = -\frac{N}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum_{n=1}^{N}(y_n - x_n^T\theta)^2 = -\frac{N}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|Y - X\theta\|^2.$$

Maximizing over θ is equivalent to minimizing $\|Y - X\theta\|^2$. Setting the gradient to zero:

$$\nabla_\theta \|Y - X\theta\|^2 = -2X^T(Y - X\theta) = 0 \implies X^TX\theta = X^TY \implies \hat{\theta} = (X^TX)^{-1}X^TY.$$

(d) As shown in (c), the log-likelihood equals $-\frac{1}{2\sigma^2}\|Y - X\theta\|^2$ plus a constant independent of θ. Thus maximizing $\ell(\theta)$ is equivalent to minimizing $\|Y - X\theta\|_2^2$.
The connection: least-squares regression corresponds to maximum likelihood estimation **when the noise is i.i.d. Gaussian** with constant variance. If the noise followed a different distribution (e.g. Laplace), the MLE would lead to a different loss (e.g. L_1 loss).

Exercise 6.2 [Hard]

Let $X = (X_1, \ldots, X_d)$ be a random vector with $X_i \sim \text{Uniform}(0, 1)$ independently.

(a) Compute $\mathbb{E}[X_i]$ and $\text{Var}(X_i)$ for each component.

(b) Let $S = \sum_{i=1}^{d} X_i$. Using the properties of expectation and variance for independent random variables, find $\mathbb{E}[S]$ and $\text{Var}(S)$.

(c) Apply the Chebyshev inequality to bound $P\big(|S - \mathbb{E}[S]| \geq t\big)$ for any $t > 0$.

(d) For what value of d can you guarantee, via Chebyshev, that $P\big(\left|\frac{S}{d} - \frac{1}{2}\right| \geq 0.1\big) \leq 0.05$?

Solution

(a) For $X_i \sim \text{Uniform}(0, 1)$:

$$\mathbb{E}[X_i] = \frac{1}{2}, \qquad \text{Var}(X_i) = \frac{1}{12}.$$

(b) By linearity of expectation and independence:

$$\mathbb{E}[S] = \sum_{i=1}^{d} \mathbb{E}[X_i] = \frac{d}{2}, \qquad \text{Var}(S) = \sum_{i=1}^{d} \text{Var}(X_i) = \frac{d}{12}.$$

(c) The Chebyshev inequality states $P(|S - \mathbb{E}[S]| \geq t) \leq \frac{\text{Var}(S)}{t^2}$. Thus:

$$P\left(\left|S - \frac{d}{2}\right| \geq t\right) \leq \frac{d}{12\,t^2}.$$

(d) We want $P\big(\left|\frac{S}{d} - \frac{1}{2}\right| \geq 0.1\big) \leq 0.05$. This is equivalent to $P(|S - d/2| \geq 0.1d) \leq 0.05$. Applying Chebyshev with $t = 0.1d$:

$$\frac{d/12}{(0.1d)^2} = \frac{d/12}{0.01\,d^2} = \frac{1}{0.12\,d} \leq 0.05.$$

Solving: $d \geq \frac{1}{0.006} = \frac{500}{3} \approx 166.7$. So $d \geq 167$ suffices.

Exercise 6.3 [Hard]

Consider a classification model that assigns class labels $Y \in \{1, 2, 3\}$ to inputs $x \in \mathbb{R}^d$. The class-conditional densities are Gaussian: $p(x|Y = k) = \mathcal{N}(x; \mu_k, \Sigma)$ with equal covariance Σ and equal prior probabilities $P(Y = k) = \frac{1}{3}$.

(a) Using Bayes' theorem, write the posterior $P(Y = k|x)$ and simplify using the equal prior assumption.

(b) Show that the decision boundary between classes k and l (where $P(Y = k|x) = P(Y = l|x)$) is a hyperplane. Find the equation of this hyperplane in terms of μ_k, μ_l, and Σ.

(c) Is the MAP (maximum a posteriori) decision rule the same as the MLE decision rule when priors are equal? Justify.

Solution

(a) By Bayes' theorem:

$$P(Y = k|x) = \frac{p(x|Y = k)P(Y = k)}{\sum_{l=1}^{3} p(x|Y = l)P(Y = l)}.$$

With equal priors $P(Y = k) = 1/3$, the $1/3$ cancels:

$$P(Y = k|x) = \frac{p(x|Y = k)}{\sum_{l=1}^{3} p(x|Y = l)}.$$

(b) The boundary satisfies $P(Y = k|x) = P(Y = l|x)$, which by (a) simplifies to $p(x|Y = k) = p(x|Y = l)$, i.e.

$$\mathcal{N}(x; \mu_k, \Sigma) = \mathcal{N}(x; \mu_l, \Sigma).$$

Since the normalizing constants are equal (same Σ), this reduces to equal exponents:

$$(x - \mu_k)^T \Sigma^{-1} (x - \mu_k) = (x - \mu_l)^T \Sigma^{-1} (x - \mu_l).$$

Expanding and canceling $x^T \Sigma^{-1} x$:

$$-2\mu_k^T \Sigma^{-1} x + \mu_k^T \Sigma^{-1} \mu_k = -2\mu_l^T \Sigma^{-1} x + \mu_l^T \Sigma^{-1} \mu_l.$$

Rearranging:

$$2(\mu_k - \mu_l)^T \Sigma^{-1} x = \mu_k^T \Sigma^{-1} \mu_k - \mu_l^T \Sigma^{-1} \mu_l.$$

This is of the form $w^T x = b$ with $w = 2\Sigma^{-1}(\mu_k - \mu_l)$, which defines a hyperplane.

(c) Yes. The MAP rule selects $\arg\max_k P(Y = k|x) = \arg\max_k p(x|Y = k)P(Y = k)$. With equal priors, this becomes $\arg\max_k p(x|Y = k)$, which is exactly the MLE decision rule (choosing the class whose model assigns the highest likelihood to x). Equal priors make MAP and MLE equivalent.